# THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

## CLASS IX — MATHEMATICS · FULL SYLLABUS EXAMINATION

Session 2026-27 | CBSE / RBSE Pattern

| Class: 9 | Subject: Maths | Set 03 | Time: 3 Hours | Max. Marks: 80 |
|---|---|---|---|---|

**General Instructions:**

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take $\pi = 22/7$ wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

### SECTION A — Objective Type — 20 questions · 20 marks

*Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.*

**1.** The point on the y-axis, 5 units below the x-axis, is: **1**

(a) (0, 5) (b) (5, 0) (c) (−5, 0) (d) (0, −5)

**2.** The point (−3, −8) lies in the: **1**

(a) second quadrant (b) first quadrant
(c) third quadrant (d) fourth quadrant

**3.** The distance of the point (−5, 12) from the origin is: **1**

(a) 17 units (b) 7 units (c) √119 units (d) 13 units

**4.** The degree of the polynomial $2x^5 - x^3 + 9$ is: **1**

(a) 3 (b) 5 (c) 2 (d) 9

**5.** The length and breadth of a rectangle are $(x + 4)$ and $(x - 1)$ respectively. Its area is: **1**

(a) $x^2 + 3x - 4$ (b) $x^2 + 5x - 4$ (c) $2x + 6$ (d) $x^2 - 4$

**6.** A linear polynomial $p(x)$ satisfies $p(0) = 3$ and $p(2) = 7$. Then $p(x)$ is: **1**

(a) $2x + 3$ (b) $x + 5$ (c) $2x - 3$ (d) $3x + 2$

**7.** Which of the following is an irrational number? **1**

(a) $\frac{22}{7}$ (b) $\sqrt{1.69}$
(c) $\frac{\sqrt{2}}{2}$ (d) 0.333… (3 repeating)

**8.** The value of $|-3.5| + |2.5|$ is: **1**

(a) 6 (b) −1 (c) 1 (d) −6

**9.** Which of the following rational numbers lies between $\frac{3}{5}$ and $\frac{4}{5}$? **1**

(a) $\frac{7}{10}$ (b) $\frac{9}{10}$ (c) $\frac{4}{5}$ (d) $\frac{1}{2}$

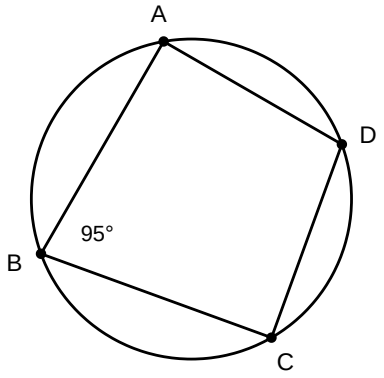
**10.** The factorisation of $x^2 + 10x + 25$ is: **1**

(a) $(x + 5)^2$ (b) $(x + 25)(x + 1)$
(c) $(x + 5)(x - 5)$ (d) $(x - 5)^2$

**11.** If $x + y = 8$ and $xy = 12$, then $x^2 + y^2$ is equal to: **1**

(a) 20 (b) 52 (c) 40 (d) 64

**12.** In the figure ABCD is a cyclic quadrilateral with $\angle ABC = 95°$. Then $\angle ADC$ is: **1**




(a) 75° (b) 85° (c) 95° (d) 170°

**13.** A chord of a circle of radius 13 cm is at a distance of 12 cm from the centre. The length of the chord is: **1**

(a) 12 cm (b) 5 cm (c) 25 cm (d) 10 cm

**14.** The area of a triangle with sides 9 cm, 40 cm and 41 cm is: **1**

(a) 360 cm² (b) 90 cm² (c) 369 cm² (d) 180 cm²

**15.** The sides of a triangular field are in the ratio 3 : 5 : 7 and its perimeter is 150 m. The shortest side is: **1**

(a) 50 m (b) 10 m (c) 30 m (d) 45 m

**16.** A bag has 3 red, 5 blue and 2 green balls. A ball is drawn at random. The probability that it is not blue is: **1**

(a) $\frac{1}{5}$ (b) $\frac{1}{10}$ (c) $\frac{3}{10}$ (d) $\frac{1}{2}$

**17.** A number is chosen at random from the numbers 1 to 20. The probability that it is a multiple of 5 is: **1**

(a) $\frac{1}{5}$ (b) $\frac{3}{20}$ (c) $\frac{1}{20}$ (d) $\frac{1}{4}$

**18.** If $u_n = (-1)^n(3n - 2)$, then $u_4$ is: **1**

(a) 14 (b) −14 (c) −10 (d) 10

**19.** **Assertion (A):** The number 0.1010010001… (with increasing number of zeros) is an irrational number. **1**
**Reason (R):** A non-terminating and non-repeating decimal is an irrational number.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

**20.** **Assertion (A):** The polynomial $3x - 5$ has exactly one zero. **1**
**Reason (R):** A linear polynomial $ax + b$ ($a \neq 0$) has exactly one zero, which is $-b/a$.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

## SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

*Internal choice is provided in one question.*

**21.** A cab charges ₹60 for the first 2 km and ₹12 for every additional km. Find the fare for a journey of 15 km. **2**

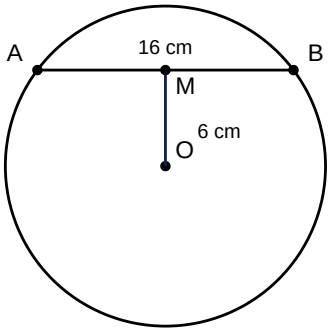
**22.** Without actual division, decide whether $\frac{27}{200}$ has a terminating decimal expansion. If it does, find the number of decimal places. **2**

**OR**

Find two rational numbers between $\frac{3}{7}$ and $\frac{4}{7}$.

**23.** Factorise: $4x^2 - 20xy + 25y^2$. **2**

**24.** In the figure, O is the centre of the circle, chord AB = 16 cm and OM ⊥ AB with OM = 6 cm. Find the radius of the circle. **2**




**25.** Find the second, third and fourth terms of the sequence defined by $t_1 = 1$ and $t_n = t_{n-1} + 2n$ for $n \geq 2$. **2**

## SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

*Internal choice is provided in two questions.*

**26.** Show that $2.142857142857\ldots = \frac{15}{7}$. **3**

**27.** A shopkeeper sold 120 shirts: **3**

| Size | S | M | L | XL |
|---|---|---|---|---|
| Shirts sold | 18 | 42 | 36 | 24 |

One shirt is chosen at random from those sold. Find the probability that it is of size (i) M, (ii) L or XL, (iii) not S.

**OR**

The number of siblings of 50 students was recorded:

| Siblings | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Students | 8 | 22 | 15 | 5 |

A student is chosen at random. Find the probability that the student has (i) exactly one sibling, (ii) at least two siblings, (iii) no sibling.

**28.** Find the coordinates of the point on the x-axis which is equidistant from A(2, −5) and B(−2, 9). **3**

**29.** Factorise: $x^3 + 6x^2 + 12x + 8$. **3**

**30.** AB and CD are equal chords of a circle with centre O. OM ⊥ AB and ON ⊥ CD. Prove that OM = ON. **3**

**31.** A triangular plot has sides 11 m, 60 m and 61 m. Find its area using Heron's formula. **3**

**OR**

The sides of a triangle are in the ratio 5 : 12 : 13 and its perimeter is 60 cm. Find its area.

## SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

*Internal choice is provided in two questions.*

**32.** The corners of a rhombus-shaped garden are A(0, 3), B(4, 0), C(0, −3) and D(−4, 0) on a grid. **5**
(a) Plot the points and join them in order. (2 marks)
(b) Find the lengths AB, BC, CD and DA. (1½ marks)
(c) Find the lengths of the diagonals and justify that ABCD is a rhombus but not a square. (1½ marks)

**33.** Find how many natural numbers between 100 and 300 are multiples of 7, and find their sum. (2½ + 2½) **5**

**34.** In a survey of 400 families, the number of vehicles owned was recorded: **5**

| Vehicles | 0 | 1 | 2 | 3 or more |
|---|---|---|---|---|
| Families | 60 | 180 | 120 | 40 |

A family is chosen at random. Find the probability that it owns (i) no vehicle (1), (ii) exactly 2 vehicles (2), (iii) at least one vehicle (2).

**OR**

A box contains 5 green, 7 yellow and 8 purple candies. One candy is drawn at random. Find the probability that it is (i) green (1), (ii) yellow or purple (2), (iii) neither green nor yellow (2).

**35.** The length of a rectangular garden is 3 m more than twice its breadth, and its area is 65 m². Form an equation in the breadth x, solve it by factorisation and find the perimeter of the garden. **5**

**OR**

A student arranges 2 square tiles of side x units, 7 rectangular strips (x units by 1 unit) and 6 small square tiles (1 unit by 1 unit) to form one rectangle. Find its length and breadth in terms of x and its perimeter when x = 3.

## SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

*Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.*

**36.** A gym charges a one-time joining fee of ₹500 and ₹300 per month. The total amount paid (in ₹) after m months is C = 500 + 300m.

(i) What is the total amount paid after 4 months? 1

(ii) State the degree of C = 500 + 300m and its constant term. 1

(iii) After how many months will the total amount paid be ₹3500? 2

**OR**

(iii) Find the total amount after 7 months. Does the situation show linear growth or linear decay? Why?

**37.** The area of a rectangular tabletop in a school workshop is given by the polynomial $x^2 + 9x + 20$ square units, where x is a positive number.

(i) Factorise $x^2 + 9x + 20$. 1

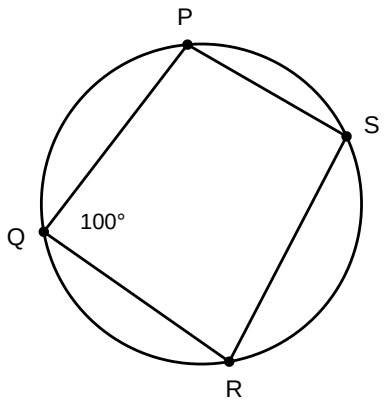
(ii) If the breadth of the tabletop is (x + 4) units, write its length. 1

(iii) Find the perimeter of the tabletop when x = 3. 2

**OR**

(iii) Using the polynomial, find the area when x = 10. Verify with (x + 4)(x + 5).

**38.** A circular running track has four floodlight poles P, Q, R and S on its boundary, forming a cyclic quadrilateral PQRS. ∠PQR = 100° and ∠QPS = 85°. The diameter of the track is 50 m.




(i) Find ∠PSR. 1

(ii) Find ∠QRS. 1

(iii) A straight path 40 m long is a chord of the track. Find its distance from the centre. 2

**OR**

(iii) Another straight path is a chord at a distance of 20 m from the centre. Find its length.